

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	1	of	12

NPS/002/004 – Technical Specification for 25-Amp Cut-outs for Street Furniture

1 Purpose

The purpose of this document is to detail the requirements of CE-Electric UK in relation to the products detailed within this document.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
DSS/005/011		Specification for 25-Amp Street Furniture Cut-Outs
MS-U-14		Specification for 25-Amp Street Furniture Cut-Outs
NPS/002/004	1	Specification for 25-Amp Street Furniture Cut-Outs

2. Scope

Cut-outs covered by this technical specification shall comply with the latest issue of:-

- BS 7654 - Specification For 25-Amp Street Lighting Cut-outs and
- ENATS 12-19 - Single Phase Street Lighting Cut-outs 25Amp Rated.

This document is intended to amplify and/or clarify CE Electric UK's requirements relating to these specifications with respect to: -

- 25amp Single-Phase, Single Pole SNE street lighting cutout c/w fuse carrier.
- 25amp Single-Phase, Single Pole PME street lighting cutout c/w fuse carrier.
- 25amp Single-Phase, Double Pole PME street lighting cut-out c/w fuse carrier.
- 25amp Single-Phase, Single Pole PME street lighting cut-out c/w fuse carrier and integral switch
- Replacement fuse carriers for above items.

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	2	of	12

2.1 Contents

1	Purpose.....	1
2.	Scope.....	1
2.1	Contents.....	2
3	Technical Requirements	3
3.1	Basic Parts.....	3
3.2	Dimensions	3
3.3	Incoming Phase Terminal Block	3
3.4	Incoming Neutral/Neutral Earth Block	3
3.5	Outgoing Terminals	3
3.6	Fuse Carrier	3
3.7	Terminal Block Cover	3
3.8	Gland Plate	3
3.9	Single pole cut out with Integral Double Pole Switch and Fuse Unit.....	4
4.1	External Documentation	5
4.2	Internal Documentation	5
4.3	Amendments from Previous Version	5
	Appendix 1 – Requirements.....	7
	Appendix 2 - Addendum to Supplier Requirement.....	8
	Appendix 3 - Logistical requirements	9
	Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION	10

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	3	of	12

3 Technical Requirements

3.1 Basic Parts

Having the ability to loop in / out with either 2 x 25mm² solid aluminium cored cables or 2 x 16mm² stranded copper cored cables; the unit shall also comprise of three basic parts. A rear panel, front terminal cover and fuse carrier. The one-piece rear panel shall comprise of all the parts necessary for the connection to the system.

3.2 Dimensions

The unit shall be capable of being fixed to a 12mm thick x 65mm wide (minimum) mounting board, by a minimum of 2 x corrosion-proof, cross-headed screws. Suitable screws should be supplied with the unit.

3.3 Incoming Phase Terminal Block

A double, incoming phase terminal block shall be formed from solid brass and electroplated. All terminal bores shall be serrated to ensure a good contact and suitable for the connection of either 2 x 16mm² stranded copper or 2 x 25mm² solid aluminium conductors.

3.4 Incoming Neutral/Neutral Earth Block

The combined Neutral/Earth or the separate Neutral and Earth blocks shall be formed from solid brass and electroplated. Terminal bores shall be serrated to ensure good contact and suitable for copper or aluminium conductors.

An insulated barrier shall be provided between any adjacent terminal assemblies.

3.5 Outgoing Terminals

The outgoing phase, neutral and earth terminals shall be suitable for terminating up to 10mm² cross-sectioned copper conductors.

3.6 Fuse Carrier

Fuse carrier shall capable of carrying a fuse-link to BS 88 with off set tag contacts and a current rating up to 25A.

It should also be capable of carrying currents associated with the operation under fault conditions.

It shall not be possible to fit the fuse-link in the neutral position.

The fuse carrier shall have a captive screw and a sealing wire facility.

A fuse link is **NOT** required to be supplied with fuse carrier / cut-out.

3.7 Terminal Block Cover

The terminal block cover shall be moulded in a clear (see through) thermoplastic material. It shall be so interlocked that it cannot be removed until the fuse-carrier has been withdrawn.

Small access holes shall be provided adjacent to the in-coming phase and neutral contact slots for testing purposes.

3.8 Gland Plate

The base gland plate shall be split plastic and be suitable for accepting up to 2 x 16mm² or 2 x 25mm² cables, each entry hole shall be fitted with an appropriate CONE shaped rubber grommets.

Document reference		NPS / 002 / 004						
Version:-	2	Date of Issue:-	Feb 2010	Page	4	of	12	

3.9 Single pole cut out with Integral Double Pole Switch and Fuse Unit

The unit shall comprise of a single-pole incoming fuse unit with terminal blocks as described within this specification. The outgoing terminals shall supply a 17.5mm din-rail mounted double-pole switch unit of rating 32 Amps. The outgoing live terminal of the double pole switch shall supply a 17.5mm din-rail cartridge fuse of rating upto 25amps.

Document reference		NPS / 002 / 004						
Version:-	2	Date of Issue:-	Feb 2010	Page	5	of	12	

4 References

4.1 External Documentation

The cut outs shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

Reference	Title
BS 7654: 1997	Specification for single phase street lighting fuses (cut-outs) for low-voltage public electricity distribution systems. 25 A rating for highway power supplies and street furniture.
ENATS 12-19: 1982	Single Phase Street Lighting Cut Outs 25Amp Rated.
BS 88	240 Volt Street lighting Cut-out Fuse Links.

4.2 Internal Documentation

Reference	Title
Nil	

4.3 Amendments from Previous Version

Clause	Subject	Amendments
3.1	Basic Parts	The inclusion for the use of 16mm Stranded Copper.
3.3	Incoming Phase Terminal Block	The inclusion for the use of 16mm Stranded Copper.
3.7	Terminal Block Cover	The terminal block cover shall be moulded in a clear (see through) thermoplastic material.
Appendix 4	Self Certification	The inclusion of self certification conformance declaration forms.

5 Definitions

Term	Definition
NIL	

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	6	of	12

6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	22/01/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	01/02/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	29/01/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	15/02/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	25/01/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	08/02/2010

Document reference		NPS / 002 / 004						
Version:-	2	Date of Issue:-	Feb 2010	Page	7	of	12	

Appendix 1 – Requirements

Requirements	CE Electric UK Commodity Code
25-Amp Single-Phase, Single Pole Separate Neutral and Earth (SNE) Cut-out c/w Fuse Carrier	044198
25-Amp Single-Phase, Single Pole Protective Multiple Earthing (PME) Cut-out c/w Fuse Carrier	044206
25-Amp Single-Phase, Double Pole Protective Multiple Earthing (PME) Cut-out c/w Fuse Carrier	181140
25-Amp Single-Phase, Single Pole Protective Multiple Earthing (PME) Cut-out c/w Fuse Carrier and Integral Switch	044230
Replacement Fuse Carriers For Above	T.B.A

Document reference		NPS / 002 / 004						
Version:-	2	Date of Issue:-	Feb 2010	Page	8	of	12	

Appendix 2 - Addendum to Supplier Requirement

Supporting evidence of compliance with type tests shall be submitted with the completed tender document.

Manufacturers may provide alternative tenders for items not complying with the above specification. This shall be clearly stated together with detailed descriptions of any variation from the specification, together with drawings and test results.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	9	of	12

Appendix 3 - Logistical requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.

Document reference		NPS / 002 / 004						
Version:-	2	Date of Issue:-	Feb 2010	Page	10	of	12	

Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

25-Amp street furniture cut-outs shall comply with the latest issues of the relevant national and international standards, including BS 7654 and ENATS 12-19. Additionally this technical specification is intended to amplify and/or clarify requirements relating to these Standards.

This self declaration sheet identifies the clauses of the aforementioned standards relevant to 25-Amp street furniture cut outs for use on the CE Electric UK distribution network. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = the product conforms fully with the requirements of this clause

Cs2 = the product conforms partially with the requirements of this clause

Cs3 = the product does not conform to the requirements of this clause

Cs4 = the product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

**Details of the cable type(s)
designed to be used with:**

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	11	of	12

Appendix 4

Technical Specification for 25Amp Street Furniture Cut Outs

• <u>ENATS 12-19</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
6.1	Non-metallic parts		
6.1.1	Interior		
6.2	Cut-out bases		
6.4	Fuse carriers		
6.4.1	Fuse contacts		
9 9.1 – 9.11	Type approval tests		

Document reference		NPS / 002 / 004					
Version:-	2	Date of Issue:-	Feb 2010	Page	12	of	12

- **BS7654**

Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.3.2.4.101	Fuse units		
7.1	Constructional requirements		
7.1.1	Materials		
7.1.2.101	Fuse carriers		
7.1.3	Clearances and creepage distances		
7.1.7	Terminals		
7.1.7.1	Constructional requirements		
7.1.7.1.1	Pillar type terminals		
7.2.2	Temperature rise		
8	Tests		
8.1.2	Type tests		
8.3.3.3.4	Temperature rise of the fuse unit		
8.3.4	Test for mechanical strength of the electrical components		